

Metadata – Dadant Hive – Series « 2024_12_23_Csv »

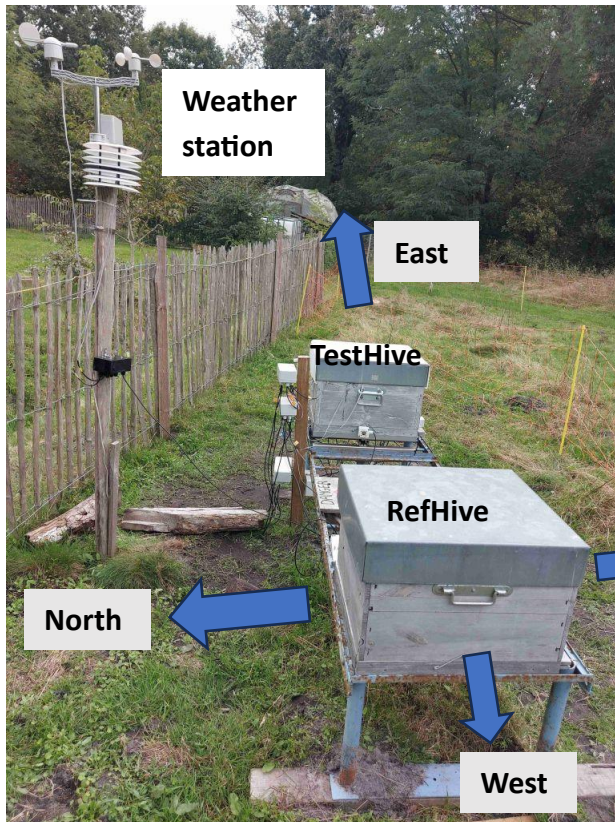
Datasets : 2024_12_23_Env.csv, 2024_12_23_RefHive.csv, 2024_12_23_TestHive.csv

Apiary location: Beguey, 33130, Saint Symphorien, France

Hive type: Empty wooden Dadant 10 frames (without frames, without bees).

Hive environment: open area

Hives orientation: Bees entrance toward south.



A wooden Dadant hive remains unmodified during the experiments to be used as a reference and is therefore called the reference hive (RefHive). Another Dadant hive, called “TestHive”, underwent several successive modifications. Since all hives are empty in this experiment, results are reproducible.

Description:

Test Hive:

- 5 sensors installed on hive roof: north, south, east, west and central position
- 5 sensors installed on overframe: north, south, east, west and central position
- 1 sensor in upper hive body
- 1 sensor in lower hive body
- 1 sensor below hive
- 2 infrared cameras

Reference hive:

- 1 sensor installed on hive roof (central position)
- 1 sensor installed on overframe (central position)
- 1 sensor in upper hive body
- 1 sensor in lower hive body
- 1 sensor below hive



Figure 1: Temperature sensor locations

Environment:

- Luminosity is measured on each of the 6 external sides of the hive (**6 sensors**).
It is not used to measure solar irradiance but rather as a qualitative measurement of luminosity (rough measurement of cloud coverage).
- Outside air sensor
- Wind speed

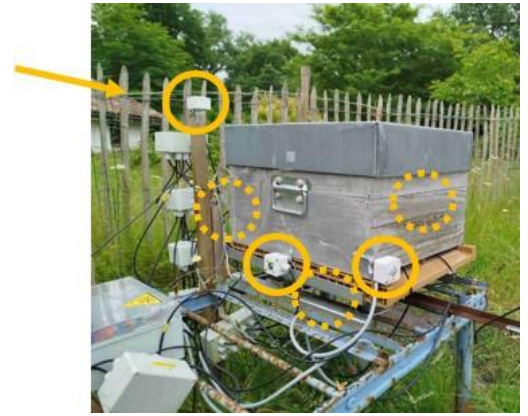


Figure 2: Luminosity sensors

Communications between the acquisition system and the sensors are done via the I2C protocol. If necessary, I2C hubs were inserted to extend the number of sensors that can be connected simultaneously.

The sensors provide an internal analog-to-digital conversion. The sensors were individually calibrated to cancel the bias error between them to allow more precise temperature difference measurements ($<0.5^{\circ}\text{C}$). Absolute temperature errors are probably around 1°C . The intercalibration were achieved by putting all sensors together in an insulated box.

The acquisition period is set to one minute except for the wind speed data which are acquired every 2 s and averaged over 1 min.

Data are stored using a csv format in 3 different files:

- 1) 2024_12_23_Env.csv
- 2) 2024_12_23_RefHive.csv
- 3) 2024_12_23_TestHive.csv

Comment lines are prefixed with hash character '#'. Column titles are built using comma separated identifier:

identifier1[.identifier2[.identifier3...]].field

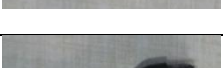
For example, the roof temperature of test hive is "test.roof.T".



Figure 3: I2C hubs

List of columns in csv file:

Field name	Description	Unit	Sensor type	Sensor location	
Timestamp	Timestamp of measurement: Year/month/ Day/Hour/ min/sec	(local time)	PCF8523 (Real-time clock)+ DS3231 (Real-time clock module)	Acquisition system	
Rtc.epoch [day]	Timestamp of measurement	Day since 1 January 0			
Air2.T [deg]	Outside air temperature	°C	NCT75	Close to the apiary, 2m high, in a radiation shield made of 5 plastic plates to limit temperature errors due to radiation heat transfer.	
Air1.T [deg]	Outside air temperature	°C	HIH8100		
Air1.HR [%]	Outside air humidity	%	HIH8100		
Wind.speed [m/s]	Wind speed	m/s	Sparkfun meteo system	Close to the apiary, 2m high	
luminosity.north.h.white [lux]	Luminosity on the north face of the hive	lux	VEML7700 (white channel)	North side of the hive	
luminosity.south.h.white [lux]	Luminosity on the south face of the hive	Lux	VEML7700 (white channel)	South side of the hive	
luminosity.east.white [lux]	Luminosity on the east face of the hive	Lux	VEML7700 (white channel)	East side of the hive	
luminosity.west.white [lux]	Luminosity on the west face of the hive	Lux	VEML7700 (white channel)	West side of the hive	
luminosity.top.white [lux]	Luminosity on the top of the hive	Lux	VEML7700 (white channel)	Top of the hive	
luminosity.below.white [lux]	Luminosity below the hive	Lux	VEML7700 (white channel)	Below the hive	
roof [deg]	Temperature of hive roof	°C	TC74	Glued under the hive roof, centered	
overframe [deg]	Temperature in hive overframe	°C	NCT75	Center of the overframe	
body.top.T [deg] body.top.HR [%]	Temperature/humidity in hive body	°C	HIH8100	Inside hive body (top), in the center	
body.bottom.T [deg] body.bottom.HR [%]	Temperature/humidity in hive body	°C	HIH8100	Inside hive body (bottom), in the center	

below [deg]	Temperature below the hive	°C	NCT75	Centred below the hive floor	
roof.north.T [deg]	Temperature of the north side of the hive roof	°C	TC74	Hive roof north	
roof.south.T [deg]	Temperature of the south side of the hiveroof	°C	TC74	Hive roof south	
roof.east.T [deg]	Temperature of the east side of the hive roof	°C	TC74	Hive roof east	
roof.west.T [deg]	Temperature of the west side of the hive roof	°C	TC74	Hive roof west	
overframe.north.T [deg]	Temperature of the north side of the overframe	°C	NCT75	Overframe north	
overframe.south.T [deg]	Temperature of the south side of the overframe	°C	NCT75	Overframe south	
overframe.east.T [deg]	Temperature of the east side of the overframe	°C	NCT75	Overframe east	
overframe.west.T [deg]	Temperature of the west side of the overframe	°C	NCT75	Overframe west	
camera.ext.south.PTAT.T [deg]	Temperature of infrared camera sensor	°C	D6T44	Internal sensor of the D6T	
camera.ext.south.T [deg]	Outside surface temperature of south side	°C	D6T44	40 cm from the south surface	
camera.ext.north.PTAT.T [deg]	Temperature of infrared camera sensor	°C	D6T44	Internal sensor of the D6T	
camera.ext.north.T [deg]	Outside surface temperature of north side	°C	D6T44	40 cm from the south surface	

For clarity, the hive names “Test” and “Ref” are omitted. As a result, “overframe [deg]” must be read as “Test.overframe [deg]” or “Ref.overframe [deg]”.

D6T44 sensors are inserted in a watertight enclosure. The window is based on a protective film (KUBE PIR Transmissive Window Films (2057, opaque, thickness. 0.14mm, 81% IR transmission) which allows most infrared radiation reaching the sensor.